JAVA INTERFACE INHERITANCE PROGRAM

Application: Online Shopping | Program 6

1. Aim

To write a Java program using interface inheritance where the ElectronicProduct interface extends the Product interface and the Laptop class implements ElectronicProduct.

2. Steps for Execution

1. Open VS Code, Eclipse or IntelliJ IDEA with JDK installed.
2. Create a Java file named **Program6.java**.
3. Enter the Java source code and save the file.
4. Compile and run the program.
5. Create the Product interface with showProduct().
6. Create ElectronicProduct extending Product and add showPrice().
7. Create the Laptop class implementing ElectronicProduct.
8. Assign the product name and price.
9. Create a Laptop object and call showProduct() and showPrice().

3. Algorithm

1. Start the program.
2. Create the Product interface with showProduct().
3. Create ElectronicProduct extending Product.
4. Declare showPrice() in ElectronicProduct.
5. Create Laptop implementing ElectronicProduct.
6. Declare the product name as Laptop and price as 50000.
7. Implement showProduct() and showPrice().
8. Create a Laptop object in main().
9. Call showProduct() and showPrice().
10. Stop the program.

4. Explanation of the Program

Product interface: It declares the showProduct() method for displaying product information.

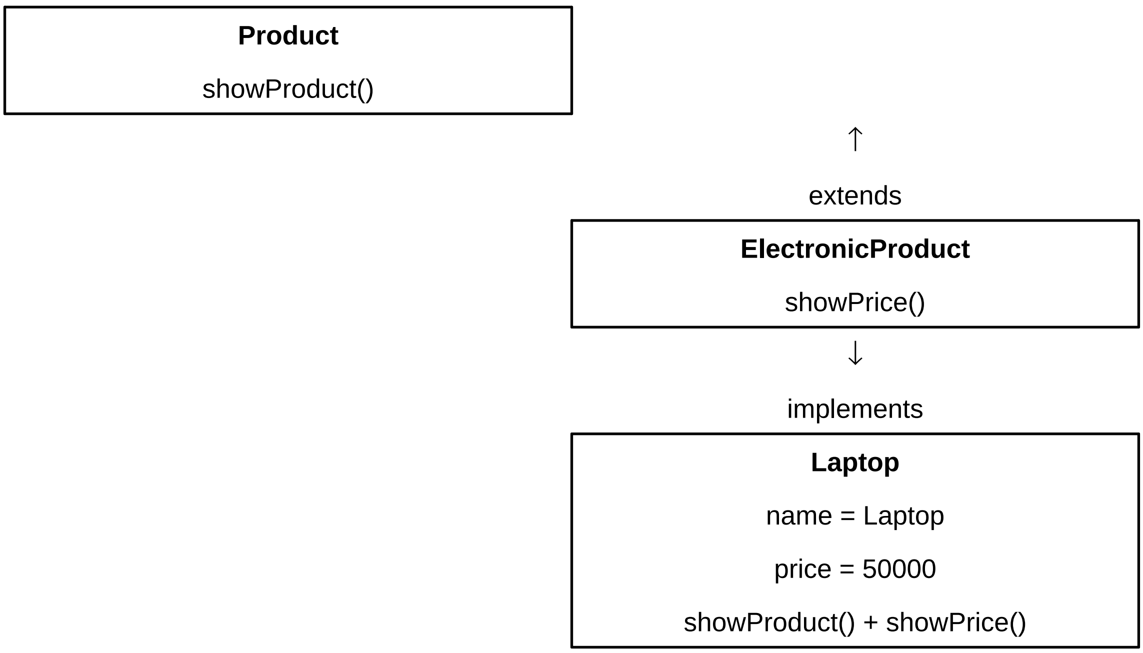
ElectronicProduct interface: It extends Product and adds the showPrice() method.

Laptop class: It implements ElectronicProduct, so it provides implementations for both showProduct() and showPrice().

Interface inheritance: ElectronicProduct inherits the showProduct() method from Product through the extends keyword.

Laptop object: The statement Laptop l = new Laptop(); creates an object used to display the product name and price.

5. Interface Inheritance Diagram



6. Java Source Code

```
// Program 6: I1 extends I2
// Application: Online Shopping

interface Product {
    void showProduct();
}

interface ElectronicProduct extends Product {
    void showPrice();
}

class Laptop implements ElectronicProduct {

    String name = "Laptop";
    int price = 50000;

    public void showProduct() {
        System.out.println("Product = " + name);
    }
```

```java
    public void showPrice() {
        System.out.println("Price = " + price);
    }
}

public class Program6 {
    public static void main(String[] args) {

        Laptop l = new Laptop();

        l.showProduct();
        l.showPrice();
    }
}
```

7. Output of the Program

```
Product = Laptop
Price = 50000
```

8. Result

Thus, the Java program successfully demonstrates interface inheritance by extending the ElectronicProduct interface from Product and implementing it in the Laptop class.